Mason Todd Prather

XR/VR Developer | Software Engineer | Research Engineer | Technical Visualization Developer

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Professional Summary

XR/VR developer and software engineer building real-time 3D applications, collaborative VR systems, simulation environments, and technical visualization tools. Experienced delivering Unity and Unreal Engine systems for Meta Quest Pro, with Python/C#/C++ development across avatar, telemetry, media, and digital-twin workflows. Focused on building practical immersive systems that translate complex workflows, spatial data, and research requirements into usable interactive experiences.

Technical Skills

XR / Real-Time 3D: Unreal Engine, Unity, OpenXR, Meta XR SDK, Meta Quest Pro, VR, collaborative VR, multi-user interaction, spatial UI, eye tracking, facial tracking, full-body avatars, technical visualization

Programming: C#, C++, Python, C, JavaScript, SQL, Bash

Systems / Data / Infrastructure: Linux, Docker, Git, P4, Tofino, InfluxDB, Grafana, PCAP workflows, JSON pipelines, REST/WebSocket integration, Cloudflare R2 upload workflows

3D / Visualization Tools: NVIDIA Omniverse, Isaac Sim, CloudXR, ROS 2, Blender, GIMP, Ready Player Me, Final IK

Research / Delivery: QA testing, participant-facing build delivery, test plans, technical documentation, troubleshooting guides, handoff documentation

Professional Experience

Research Engineer — Feb. 2024 – Present

OPEX Systems — Marietta, GA

- Engineer applied software prototypes for programmable-network telemetry, intrusion-detection testbeds, digital-twin visualization, and immersive engineering workflows.
- Build Python control-plane tooling for P4/Tofino environments, including digest processing, register reads, socket messaging, flow lifecycle tracking, and structured data export.
- Integrate flow metrics and event data with InfluxDB/Grafana dashboards for monitoring, classifier validation, testbed analysis, and engineering demonstrations.
- Support VR and digital-twin visualization pipelines using NVIDIA Omniverse, Isaac Sim, CloudXR, ROS 2, and Quest headset streaming.
- Author test plans, troubleshooting guides, implementation notes, and handoff documentation that make complex prototypes easier to validate, demo, and maintain.

Graduate Research Assistant / XR Developer — Aug. 2024 – May 2026

Kennesaw State University, Department of Software Engineering and Game Development — Kennesaw, GA

- Developed Meta Quest Pro VR systems across collaborative storytelling, therapy-oriented social interaction, medical visualization, and study-support tooling.
- Contributed XR implementation to IEEE VR 2025 and IEEE VR 2026 3DUI Contest publication/entry projects involving multi-scale World-in-Miniature interaction, multi-user VR, and multi-modal presentation workflows.
- Built Wind Phone Therapy features for bereaved-parent support scenarios, including full-body avatars, eye/facial tracking, avatar expression mapping, phone-based media upload/sharing, and XR scene composition.
- Designed an automated VR study video pipeline for a Pennsylvania State University-led workplace simulation study, capturing participant headset perspective, recording through completion, finalizing video files, and uploading to Cloudflare R2 through a custom web API.
- Delivered participant-ready builds, documented technical workflows, and collaborated with faculty and research partners to align XR implementation with project goals.

Undergraduate Research Assistant / XR Developer — Apr. 2022 – Aug. 2024

Kennesaw State University, Realities Lab — Kennesaw, GA

- Developed Unity VR applications for faculty-led research involving immersive training, interaction studies, and educational/serious-game prototypes.
- Implemented gameplay scripts, UI flows, input handling, scene setup, QA testing, deployment support, and participant-facing builds.
- Translated study requirements into working XR features while debugging user flows, headset interactions, and scene behavior with faculty and lab collaborators.

- Supported published and presented research work in eye-tracking attention training, VR interaction studies, and public research demonstrations.

Graphic Designer Sept. 2019 – Jan. 2021
Helix Gaming Remote / Esports Organization

- Produced esports graphics, branding assets, social media visuals, and event-facing content for competitive gaming communities.

Selected Technical Projects

Wind Phone Therapy VR System KSU Graduate Research
Meta Quest Pro, avatars, eye/facial tracking, media sharing

- Built social-interaction and media-sharing features for therapy-oriented VR scenes, emphasizing expressive avatars and respectful immersive composition.

IEEE VR 2025 / 2026 3DUI Contest Projects KSU Graduate Research
Collaborative VR, medical visualization, multi-user VR, multi-modal interaction

- Supported contest-entry VR systems that translated narrative, spatial navigation, and medical-imaging presentation concepts into usable immersive interactions.

VR Study Video Recording & Cloud Upload Pipeline Penn State-led Study Support
Meta Quest Pro, video processing, Cloudflare R2, custom web API

- Created a hands-off workflow for participant-view recording, study-completion capture, finalized video handling, and cloud upload for review.

Network Telemetry & Intrusion Detection Testbed OPEX Systems
P4, Python, Tofino, InfluxDB, Grafana, PCAP workflows

- Developed monitoring and validation workflows for flow telemetry, traffic replay, classifier support, and programmable-switch experiments.

CloudXR / Omniverse Digital Twin Workflow OPEX Systems
NVIDIA Omniverse, Isaac Sim, CloudXR, ROS 2, Quest 3

- Supported containerized rendering and VR streaming workflows for interacting with technical 3D scenes through headset-based visualization.

Education

Kennesaw State University May 2026
Master of Science in Software Engineering Kennesaw, GA

- Thesis focus: immersive VR learning systems, attention timing, and learner-experience tradeoffs.

Kennesaw State University May 2024
Bachelor of Science in Computer Game Design and Development Kennesaw, GA

Publications, Presentations & Honors

- IEEE VR 2025 3DUI Contest publication/entry: *Saving the Polar Bears: A Collaborative VR Storytelling Experience Enhanced by Multi-scale and Multi-view WiM Designs.*
- IEEE VR 2026 3DUI Contest publication/entry: *Beyond Slices: A Narrative-Driven, Multi-User, Multi-Modal Virtual Reality System for Medical Imaging Presentation.*
- ISMAR 2025 research publication connected to XR study work involving *Scooter Coin.*
- Dean's List, Spring 2021 and Fall 2024, Kennesaw State University; 1st Place, Undergraduate Capstone Projects, KSU C-Day 2022.
- Presented XR/game development work at academic and public-facing research events, including Atlanta Science Festival 2024.